Q.1 Investigate\,\,the\,\,\lim it\,\,of\,\,\,\mathop {\lim }\limits_{x \to 0} \,\,x\sin \frac{{1\,\,}}{x} = - - - - .

A. none\,\,of\,\,these

B. 1

C. 0

D. x

Q.2 Investigate\,\,the\,\,\lim it\,\,of\,\,\,\mathop {\lim }\limits_{x \to 0^ + } \,\,\left({\frac{{\left| x \right|}}{x} + x} \right) = - - - - .

A. 0

B. none\,\,of\,\,these

C. 1\,

D. - 1

Q.3 Rolle's Theorem says, suppose that f is continuous on the closed interval [a,b] and differentiable on the open interval (a,b) and f(a) = f(b).

A. Then f'(c) = 0 for some c in the open interval (a,).

B. Then f'(c) exists for some c in the open interval (a,b).

C. Then f'(c) 0 for some c in the open interval (a,b).

D. Then f'(c) = 0 for some c in the open interval (a,b).

Q.4 \begin{array}{l} The\,series\,\sum\limits_{}^{} {a_n b_n } \,\,\,converges\,\,if\,\,a_{n + 1} \le \,a_{n\,} \,for\,\,n\, - - - k,\,\,\mathop {\lim }\limits_{x \to \infty } \,\,a_n = 0,\,\,and\, \\\ |b_k + \,b_{k + 1} + ... + b_n | \le M,\,\,for\,\,some\,\,constant\,M. \\\ \end{array}

A. <

B. >

C. \,

D. \,

Q.5 \begin{array}{l} The\,series\,\sum\limits_{}^{} {a_n b_n } \,\,\,converges\,\,if\,\,a_{n + 1} \le \,a_{n\,} \,for\,\,n\, \ge \,k,\,\,\mathop {\lim }\limits_{x \to \infty } \,\,a_n = - - - - - ,\,\,and\, \\\ |b_k + \,b_{k + 1} + ... + b_n |\,\, \le \,\,M,\,\,for\,\,some\,\,constant\,M. \\\ \end{array}

A. 1

B. 0\,

C. None\,of\,these

D. - 1

Q.6 Which function has 0/0 form

A. f(x) = 2 x at x=2

B. f(x) = x + x at x =0

C. f(x) = { x}{x-1} at x=1

D. f(x) = {e^{-x}}{x} at x=2

Q.7 If f is differentiable at a local extreme point x_0\in D_{f}^{0}, then

A. |f'(x_0)|<~0.

B. f'(x_0) ~0.

C. f'(x_0)=~0.

D. x_0=~0.

Q.8 If f(x) = |x| and g(x) = |5x – 2|. Then, fog =

A. f(x)

B. |g(x)|

C. |f(x)|

D. f^2(x)

Q.9 The\,series\,\sum\limits_{}^{} {(- 1)^n a_n } \,\,\, - - - - - - \,\,\,if\,\,0 \le a_{n + 1} \le \,a_{n\,} \,and\,\mathop {\lim }\limits_{x \to \infty } \,\,a_n = 0

A. diverges

B. converges

Q.10 For two functions f(x) and g(x), the composit function fog(x) is defined as -------

A. None of these

B. (f-g)(x)

C. g(f(x))

D. f(g(x))
